

CERVICAL CANCER CONTROL EFFORTS IN NIGERIA: A CONTENT EVALUATION OF THE NATIONAL CANCER CONTROL PLAN FOR NIGERIA (2018-2022).

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EXECUTIVE SUMMARY

Cancer is a public health problem, ranking as the second leading cause of death worldwide. In Nigeria, cervical cancer poses a particularly significant threat, with an estimated 14,943 new cases and 10,403 deaths annually. Nigeria's National Cancer Control Plan, first established in 2006, aimed to reduce cancer incidence and prevalence through structured approaches and policies, and has gone through multiple iterations. The second edition (2018-2022) focused on increasing screening services, and the current third edition (2023-2027) was launched in 2022 to continue addressing the country's cancer burden. This evaluation utilised secondary data from policy documents of the NCCP and WHO and reveals that while the plan aims to reduce cancer incidence and mortality through a comprehensive strategy encompassing prevention, diagnosis, treatment, palliative care, advocacy, data management, and sustainable funding, it falls short in defining specific objectives and targets for cervical cancer prevention and early detection. This lack of clear focus may impede effective efforts to address this preventable disease. While the NCCP aligns with the WHO guidelines and recommendations for cancer control programs, the absence of detailed strategies and success indicators for cervical cancer control is a notable weakness.

The evaluation of Nigeria's NCCP (2018-2022) identifies a significant gap between the current screening practices and the plan's goal of reaching over 50% screening coverage by 2022. It also highlights the urgent need to address cervical cancer, which significantly impacts women's health in the country.

INTRODUCTION

Cancer is a global problem as it is the second leading cause of death worldwide (Uchendu 0.J.,2020 1) and unfortunately, it lacks a global solution (*JAMA Oncol.* 2019 2). It is a broad term that covers a wide variety of diseases that can start in almost any part of the body, and it involves the uncontrollable abnormal growth of cells beyond their usual boundaries (WHO 2022), invading adjoining parts of the body and spreading to other organs through a process known as metastasis (a major cause of death from cancer). Cervical cancer is the malignant neoplasm of the cervix uteri, and it is a type of cancer caused by persistent genital high-risk human papillomavirus (HPV) infection. Globally, it ranks as the fourth most common cancer, with high incidence and mortality in women (Mafiana et al., 2022), and a large share of this burden is especially pervasive on low- and middle-income countries like Nigeria (Runice et al., 2023).

Nigeria is an LMIC, and it is the largest country in Africa. It has a population of over 229 million people (Worldmeter 2024), and of all the factors associated with its cancer burden, the advanced stage at which diagnosis is carried out stands out distinctly. The burden of cancer in Nigeria is significant with an annual estimate of 102,000 new cancer cases and 72,000 cancer deaths (GLOBOCAN 2022). The incidence of cervical cancer in Nigeria is about 18.4 per 100,000 women, with an estimated 12,075 women diagnosed every year (Lawson et al 2023). There are roughly 14,943 new cases of cervical cancer being diagnosed in Nigeria, resulting in 10,403 deaths (GLOBOCAN 2022), and it is estimated that by 2025, the number of deaths caused by cervical cancer in the country will increase by 63% for women aged ≤ 65 , and by 50% for those aged > 65 years (Olubodun et al 2022).

Cervical cancer is largely preventable through HPV vaccination and early detection via screening, yet awareness and uptake of these interventions remain low in Nigeria. Although the Nigerian Government is putting effort towards tackling cancer diagnosis and treatment in the country, the incidence and mortality rates continue to increase (Morounke et al., 2017). So, as an effort to address this challenge, they developed a National Cancer Control Plan (NCCP).

NATIONAL CANCER CONTROL PLAN (NCCP) FOR NIGERIA (2018-2022)

Cancer control involves the actions taken to reduce cancer's impact in a community or nation, based on the best scientific evidence available (Eguzo et.al., 2020 12). These actions are often supported by policies that can also be called cancer control plans or frameworks, which when implemented, result in cancer control programs (Eguzo 2022 13).

The National Cancer Control Plan was Nigeria's first cancer control policy. It was established in 2006 with a vision to reduce the incidence and prevalence of cancer in the country. In 2008, it launched its first edition which was operational for 5 years (2008 – 2013) (Lawal et. Al., 2023). The vision of the NCCP for Nigeria over the years has remained the same; to

reduce the incidence and prevalence of cancer, with their goals and objectives outlined clearly, providing a structured approach for cancer control in Nigeria. After some reflections on the first edition's goals, priority areas, and strategies, the second edition which would be the focus of this evaluation, was launched on Friday, 13th April 2018 (Cancer Aware Nigeria, 2018), and lasted 5 years. It was developed to address the country's increasing cancer burden and to make screening services available to all Nigerians, reaching at least 50% of all eligible populations by 2022 (Sharma et al., 2021).

During the 2022 International Cancer Week in Abuja, the NCCP launched its third edition, which is expected to have a lifespan of 5 years and elapse in 2027 (2023-2027) (Zubairu, Balogun 2023).

The purpose of this evaluation is to assess the content of the NCCP 2018-2022, with particular attention to cervical cancer control efforts. The level of awareness of cervical cancer and its preventive strategies is low among the Nigerian population and its policymakers (Mafiana et al., 2022 4), as there are no organised screening programs, and the few available services are only provided opportunistically with little or no effect. This paper would be a valuable source for policymakers during policy reviews. Given the potential to significantly reduce cervical cancer incidence and mortality through evidence-based interventions (Canfell et al 2020 18), it is critical to evaluate how well the NCCP (2018-2022) addresses this specific cancer type.

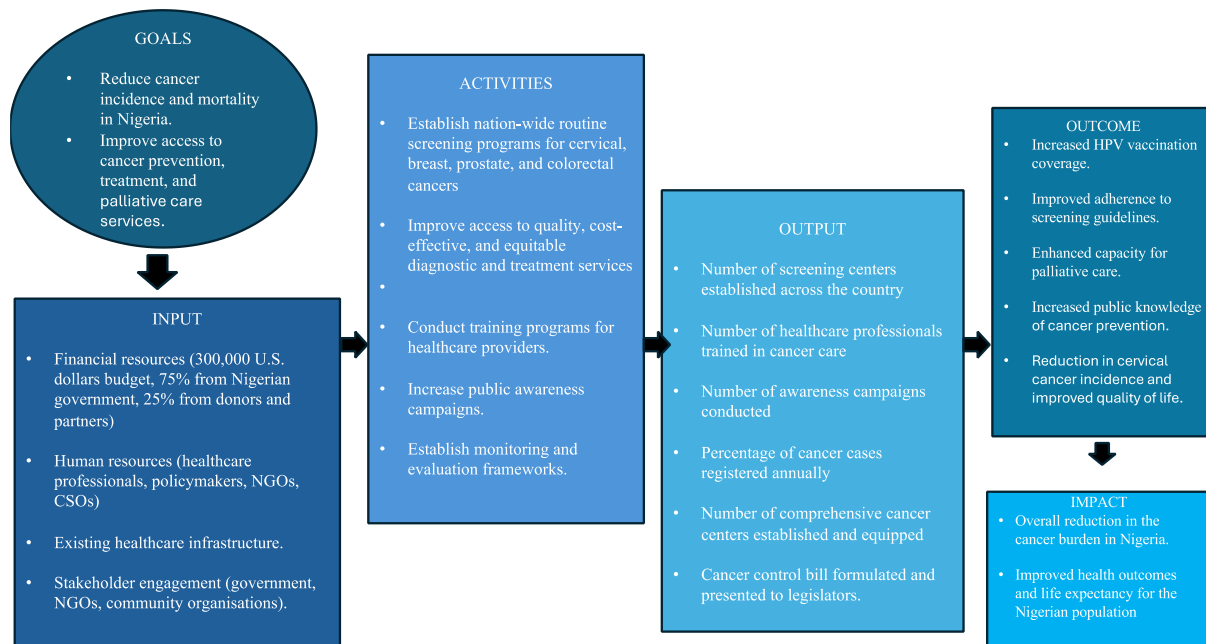
The following research questions will be assessed for this evaluation.

1. To what extent does the NCCP clearly state its goals and objectives for cervical cancer prevention and control?
2. Are the cervical cancer prevention and control strategies in the NCCP consistent with those of model policies?
3. Does the NCCP identify specific indicators for measuring the success of its cervical cancer prevention and control efforts?

METHOD:

This evaluation employs a Qualitative content analysis approach, guided by the CDC's framework for policy evaluation in public health, to evaluate the content of the NCCP (2018-2022) for Nigeria. It involved a thorough review of the NCCP policy document, extracting relevant information on its goals, objectives, strategies, and indicators. It drew upon data from literature reviews of published studies, previous editions of the NCCP, government reports, the National Library of Medicine, WHO NCCP: policies and managerial guidelines, LANCET, PUBMED, GLOBOCAN, and independent studies. Search terms like "health policy, cervical cancer, cervical cancer in Nigeria, cancer in Nigeria, public health, LMIC, NCCP Nigeria" were utilised during this evaluation. Google, Google Scholar and the University PRIMO were also used as search tools, especially for papers obtained from the reference lists of published papers. A logic model was developed to visualize the plan's goals, inputs, activities, outputs, and intended outcomes and impact.

**LOGIC MODEL FOR THE CONTENT EVALUATION OF THE NATIONAL
CANCER CONTROL PLAN (NCCP) 2018-2022 FOR NIGERIA.**



Source: NCCP for Nigeria (2018).

It is important to note that a significant challenge while evaluating the content of the NCCP (2018-2022) for Nigeria, was the lack of accurate baseline data. Most of the planning regarding the policy was based on calculated estimates and not precise incidence Zubairu et al., (2023).

FINDINGS

GOALS AND OBJECTIVES

The Federal Ministry of Health, Nigeria, established the NCCP to develop, coordinate, and implement activities, policies, and programs to address the rising incidence and mortality rates of cancer in the country, particularly among women (Jedy-Agba et al., 2012). The overall goal of the NCCP is to reduce the incidence and prevalence of cancer in Nigeria (NCCP 2018). It outlines seven priority areas of action, including

1. **PREVENTION:** Increase awareness of cancer risk factors, promote healthy lifestyles, implement vaccination programs for cancer-causing infections (e.g., HPV), and enhance early detection through screening programs.
2. **DIAGNOSIS AND TREATMENT:** Improve diagnostic capabilities, enhance treatment facilities, and ensure access to essential cancer medicines.
3. **SUPPLY CHAIN MANAGEMENT (LOGISTICS):** Ensure the availability and accessibility of essential cancer medicines and technologies.

4. HOSPICE AND PALLIATIVE CARE: Establish palliative care units and ensure quality of life for cancer patients.
5. ADVOCACY AND SOCIAL MOBILIZATION: Increase public awareness and advocacy for cancer control.
6. DATA MANAGEMENT AND RESEARCH: Support cancer registries and promote cancer research.
7. GOVERNANCE AND FINANCE: Ensure sustainable funding and effective management of cancer control programs.

Each priority area has its own set of objectives. However, specific objectives for cervical cancer control could be more clearly defined as within Nigeria's public health system, there are no officially established or systematically implemented programs focused on preventing cervical cancer, especially for women living in rural areas (Onyenwenyi and Gugu 2016). This highlights a lack of targeted efforts to address the cervical cancer prevention needs of the country through organised public health initiatives.

CONSISTENCY WITH MODEL POLICIES

The NCCP (2018) for Nigeria outlines comprehensive strategies for cancer control, focusing on various aspects of prevention, diagnosis, treatment, and palliative care. It was developed with the aim of reducing cancer incidence and mortality through a coordinated and systematic approach. WHO provides several recommendations for cancer control (2002), which typically include the following key components listed in the NCCP policy document (NCCP 2018).

Table 1 summarises the consistency between the NCCP (2018-2022) document for Nigeria and WHO recommendations for cancer control.

TABLE 1:

WHO RECOMMENDATION	NCCP (2018-2022)	CONSISTENCY
PREVENTION	PREVENTION	CONSISTENT
EARLY DETECTION	DIGNOSIS AND TREATMENT	PARTIALLY CONSISTENT
PALLIATIVE CARE	HOSPICE AND PALLIATIVE CARE	CONSISTENT
SURVEILLANCE AND RESEARCH	DATA MANAGEMENT AND RESEARCH	CONSISTENT
HEALTH SYSTEM STRENGTHENING	SUPPLY CHAIN MANAGEMENT (LOGISTICS)	PARTIALLY CONSISTENT
RESOURCE MOBILIZATION	GOVERNANCE AND FINANCE	CONSISTENT
	ADVOCACY AND SOCIAL MOBILIZATION	NCCP SPECIFIC

source: NCCP (2018), WHO (2002).

The Nigeria National Cancer Control Plan (NCCP) 2018-2022 aligns well with WHO guidelines, emphasising the prevention of cervical cancer through HPV vaccination, promoting early detection with a target of >50% screening coverage, and enhancing diagnostic and treatment capabilities, including access to essential medicines. It prioritises palliative care to improve patient's quality of life and supports evidence-based strategies through cancer registries like the Nigerian National System of Cancer Registries (NSCR). The plan addresses sustainable funding and effective management, developed with extensive cross-sectoral collaboration and broad stakeholder engagement.

SUCCESS INDICATORS

The NCCP includes a monitoring and evaluation framework with some indicators for measuring progress. It uses data obtained from population-based cancer registries to guide and measure the success of its strategies (Anyasi , Fodd 2020). However, cervical cancer-specific indicators are limited. The plan lacks clear targets for HPV vaccination coverage, cervical cancer screening rates, and treatment access. More comprehensive and specific indicators would enhance the ability to track the progress and impact of cervical cancer control efforts.

DISCUSSION

The evaluation reveals some strengths and areas for improvement in the content of Nigeria's NCCP policy document, particularly concerning cervical cancer control. The plan's comprehensive approach, addressing multiple aspects of cancer control, aligns with WHO recommendations. However, the lack of specific objectives and targets for cervical cancer prevention and early detection may hinder progress in reducing the burden of this preventable disease.

The NCCP's goal to increase cancer screening to >50% of eligible populations by 2022 is ambitious, given the low baseline reported in recent studies. For instance, Sharma et al., (2021) found that only 1.5% of eligible women reported having cervical cancer screening in a rural Nigerian community. This highlights the significant gap between current practices and the NCCP's targets, emphasising the need for more focused strategies and resource allocation. Although the plan mentions increasing cancer screening, it does not provide cervical cancer-specific targets, and the lack thereof may hinder focused efforts to prevent and detect it early. This aligns with a study, by Ginsburg et al., (2017), which emphasises that without clear objectives, resources and initiatives may not be adequately directed toward cervical cancer. This could result in insufficient screening coverage and early detection efforts.

The plan's compliance with WHO guidelines and recommendations for cancer control (2002) demonstrates Nigeria's commitment to international best practices in cancer control. However, it lacks some detailed disease-specific strategies as seen in Ghana's cancer control plan (Ghana NCCP 2011), which provides more detailed information about the objectives and strategies for specific diseases such as breast and cervical cancer (Eguzo et al., 2020). The Nigerian NCCP could benefit from more detailed implementation strategies, clearer stakeholder roles, and greater emphasis on research and disease-specific approaches while building its strength of inclusive development. Zambia's Cervical Cancer Prevention Program (CCPPZ) offers valuable lessons for Nigeria in tackling cervical cancer incidence and mortality. Nigeria could improve cervical cancer prevention by adopting its nurse-led VIAC screening approach, integrating services with HIV care, setting clear screening targets, and offering free screenings to enhance accessibility and coverage, especially in rural and low-income areas (Zambia NCCSP 2016))

The limited specificity of success indicators, especially for cervical cancer control, is a notable weakness of the NCCP as clear, measurable targets are essential for monitoring progress and guiding resource allocation. The WHO's Global Strategy to Accelerate the Elimination of Cervical Cancer (2020), provides a framework for setting specific targets, such as 90% HPV vaccination coverage, 70% screening coverage, and 90% treatment of precancerous lesions. Incorporating similar targets like that of the CCPPZ into the NCCP would increase the country's effectiveness and alignment with global elimination efforts.

RECOMMENDATIONS

1. Enhance cervical cancer-specific objectives: Develop clear, measurable targets for HPV vaccination coverage, cervical cancer screening rates, and treatment access.
2. Strengthen implementation strategies: Provide more detailed guidance on operationalising cervical cancer prevention and control programs, including community engagement and health system strengthening.
3. Improve monitoring and evaluation framework: Incorporate comprehensive indicators for cervical cancer control, aligned with WHO's elimination targets.
4. Increase resource allocation: Prioritize funding for cervical cancer prevention efforts, including HPV vaccination and screening programs.
5. Enhance collaboration: Foster partnerships with international organisations and successful national programs to leverage expertise and resources.

CONCLUSION

Nigeria's National Cancer Control Plan 2018-2022 provides a foundation for addressing the country's cancer burden, including cervical cancer. While the plan aligns with many international recommendations, there are opportunities to strengthen its approach to cervical cancer control. By enhancing specific objectives, implementation strategies, and success indicators, Nigeria can improve its efforts to reduce cervical cancer incidence and mortality. Future editions and alterations of the NCCP should prioritise these areas to ensure more effective and targeted interventions in the fight against cervical cancer.

REFERENCE

Anyasi, H.I. and Foss, A.M. (2020) 'A comparative analysis of cervical cancer prevention between Nigeria and Nordic countries that have experienced a decline in cervical cancer incidence', *International Health*, 12(1), pp. 1-10. Available at:

https://researchonline.lshtm.ac.uk/id/eprint/4657895/1/Anyasi_Foss_2020_A-comparative-analysis-of-cervical.pdf (Accessed: 27 July 2024).

Cancer Aware Nigeria, The Nigeria National Cancer Control Plan (2018-2022). Available at: <https://www.canceraware.org.ng/post/2018/04/19/the-national-cancer-control-plan>

Cancer Control: Knowledge into Action: WHO Guide for Effective Programmes: Module 5: Palliative Care. Geneva: World Health Organization; 2007. INTRODUCTION TO THE CANCER CONTROL SERIES. Available at: <https://www.ncbi.nlm.nih.gov/books/NBK195247/>

Canfell K, Kim JJ, Brisson M, Keane A, Simms KT, Caruana M, Burger EA, Martin D, Nguyen DTN, Bénard É, Sy S, Regan C, Drolet M, Gingras G, Laprise JF, Torode J, Smith MA, Fidarova E, Trapani D, Bray F, Ilbawi A, Broutet N, Hutubessy R. Mortality impact of achieving WHO cervical cancer elimination targets: a comparative modelling analysis in 78 low-income and lower-middle-income countries. *Lancet*. 2020 Feb 22;395(10224):591-603. doi: 10.1016/S0140-6736(20)30157-4. Epub 2020 Jan 30. PMID: 32007142; PMCID: PMC7043006. Available at:

[https://www.thelancet.com/journals/lancet/article/PIIS0140-6736\(20\)30157-4/fulltext](https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(20)30157-4/fulltext)

Eguzo, K., 2022. COMMUNITY ENGAGED APPROACH TO CANCER CONTROL POLICY IN ABIA STATE-NIGERIA: A MIXED-METHODS ACTION RESEARCH PROJECT (Doctoral dissertation, University of Saskatchewan). Available at:

<https://harvest.usask.ca/server/api/core/bitstreams/4a3b20b7-9676-4c8c-9a95-eb13239c3406/content>

Eguzo, K., Ramsden, V.R., Ekanem, U., Olatunbosun, O., Walker, V. and Mpofu, C., 2020. Review of cancer control policy in Nigeria and comparison with selected African countries: Implications for future policy making. *Ibom Medical Journal*, 13(1), pp.1-11. Available at: <http://ibommedicaljournal.org/index.php/imjhome/article/view/183>

Ginsburg O, Bray F, Coleman MP, Vanderpuye V, Eniu A, Kotha SR, Sarker M, Huong TT, Allemani C, Dvaladze A, Gralow J, Yeates K, Taylor C, Oomman N, Krishnan S, Sullivan R, Kombe D, Blas MM, Parham G, Kassami N, Conteh L. The global burden of women's cancers: a grand challenge in global health. *Lancet*. 2017 Feb 25;389(10071):847-860. doi: 10.1016/S0140-6736(16)31392-7. Epub 2016 Nov 1. PMID: 27814965; PMCID: PMC6191029. Available at:

[https://www.thelancet.com/journals/lancet/article/PIIS0140-6736\(16\)31392-7/abstract](https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(16)31392-7/abstract)

Global Burden of Disease Cancer Collaboration. Global, Regional, and National Cancer Incidence, Mortality, Years of Life Lost, Years Lived With Disability, and Disability-Adjusted Life-Years for 29 Cancer Groups, 1990 to 2017: A Systematic Analysis for the

Global Burden of Disease Study. JAMA Oncol. 2019;5(12):1749–1768.

doi:10.1001/jamaoncol.2019.2996 Available at:

<https://jamanetwork.com/journals/jamaoncology/fullarticle/2752381>

Global Cancer Observatory, 2022. Available at:

https://gco.iarc.fr/today/en/dataviz/bars?types=0_1&mode=cancer&cancers=23&sort_by=value1&sexes=2&populations=566&multiple_populations=0 Accessed 10th July 2024

Jedy-Agba E, Curado MP, Ogunbiyi O, Oga E, Fabowale T, Igbinoba F. Cancer Incidence in Nigeria A Report from Population-based Cancer Registries. Cancer Epidemiol.

2012;36:e271–8. doi: 10.1016/j.canep.2012.04.007. Available at:

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3438369/>

Lawson, O., Ameyan, L., Tukur, Z. *et al.* Cervical cancer screening outcomes in public health facilities in three states in Nigeria. *BMC Public Health* 23, 1688 (2023). Available at: <https://doi.org/10.1186/s12889-023-16539-1>

Lawal Qudus, O., Lawal Ishak, K. and Carter Cody, S., Cervical Cancer Burden in Nigeria: Review of Current Situation. *Gynecol Reprod Health*. 2023; 7 (1): 1-

5. Correspondence: *Qudus O. Lawal, Irrua Specialist Teaching Hospital, Irrua, Nigeria, Tel, 234(8039208185)*. Available at: <https://www.scivisionpub.com/pdfs/cervical-cancer-burden-in-nigeria-review-of-current-situation-2608.pdf>

Mafiana JJ, Dhital S, Halabia M, Wang X. Barriers to uptake of cervical cancer screening among women in Nigeria: a systematic review. *Afr Health Sci*. 2022 Jun;22(2):295-309.

doi: 10.4314/ahs.v22i2.33. PMID: 36407354; PMCID: PMC9652621. Available at:

<https://www.ajol.info/index.php/ahs/article/view/228970>

Morounke, S.G., Ayorinde, J.B., Benedict, A.O., Adedayo, F.F., Adewale, F.O., Oluwadamilare, I. and Benjamin, A., 2017. Epidemiology and incidence of common cancers in Nigeria. Population, 84(82,231,000), pp.166-629. Available at:

https://www.researchgate.net/profile/Fadaka-Adewale-Oluwaseun/publication/321700703_Epidemiology_and_Incidence_of_Common_Cancers_in_Nigeria/links/5a2bc579a6fdccfbf854fb7/Epidemiology-and-Incidence-of-Common-Cancers-in-Nigeria.pdf

NATIONAL CANCER CONTROL PLAN NIGERIA (2018-2022). 2018. [pdf] Available at: [https://www.iccp-portal.org/system/files/plans/NCCP_Final%20\[1\].pdf](https://www.iccp-portal.org/system/files/plans/NCCP_Final%20[1].pdf)

NATIONAL STRATEGY FOR CANCER CONTROL IN GHANA 2012-2016 (2011).

Available at: [https://www.iccp-](https://www.iccp-portal.org/system/files/plans/Cancer%20Plan%20Ghana%20Ministry%20of%20Health.pdf)

[portal.org/system/files/plans/Cancer%20Plan%20Ghana%20Ministry%20of%20Health.pdf](https://www.iccp-portal.org/system/files/plans/Cancer%20Plan%20Ghana%20Ministry%20of%20Health.pdf)

Olubodun T, Balogun MR, Odeyemi AK, Odukoya OO, Ogunyemi AO, Kanma-Okafor OJ, Okafor IP, Olubodun AB, Ogundele OOP, Ogunnowo B, Osibogun A. Barriers and recommendations for a cervical cancer screening program among women in low-resource settings in Lagos Nigeria: a qualitative study. *BMC Public Health*. 2022 Oct

12;22(1):1906. doi: 10.1186/s12889-022-14314-2. PMID: 36224656; PMCID: PMC9560022. Available at: <https://pubmed.ncbi.nlm.nih.gov/36224656/>
Onyenwenyi, A.O.C. and Gugu, G.M., 2016. Strategies for the prevention and control of cervical cancer in rural communities: a Nigerian perspective. *Journal of*

Community Medicine and Primary Health Care, 28(2), pp.77-93. Available at: <https://www.ajol.info/index.php/jcmphc/article/view/154699>

Runcie C W Chidebe, L Leigh Leibel, Tochukwu C Orjiakor, Ashish Shrestha, Agha A Agha, Duygu Altinok Dindar, Batuhan Kisakol, Sampson C Ipiankama, Ifeoma J Okoye, Malcolm Mason, Aviane Auguste, Darlington K Esiaka, Promoting cancer prevention through World Cancer Day in Nigeria, *The Lancet Oncology*, Volume 24, Issue 4, 2023, Pages 319-322, ISSN 1470-2045. Available at: [https://doi.org/10.1016/S1470-2045\(22\)00692-1](https://doi.org/10.1016/S1470-2045(22)00692-1)

Sharma A, Alatisie OI, O'Connell K, et al Healthcare utilisation, cancer screening and potential barriers to accessing cancer care in rural South West Nigeria: a cross-sectional study *BMJ Open* 2021;11:e040352. doi: 10.1136/bmjopen-2020-040352. Available at: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8314695/>

The REPUBLIC OF ZAMBIA, NATIONAL CANCER CONTROL STRATEGIC PLAN (2016-2021), Available at: <https://www.iccp-portal.org/system/files/plans/NCCSP%20Final%20Version%20Zambia.pdf>

Uchendu, O.J., 2020. Cancer incidence in Nigeria: a tertiary hospital experience. *Asian Pacific Journal of Cancer Care*, 5(1), pp.24-32. Available at: <http://waocp.com/journal/index.php/apjcc/article/view/335>

WHO, G.T., 2020. Global strategy to accelerate the elimination of cervical cancer as a public health problem. Available at: <https://www.who.int/publications/i/item/9789240014107>

World Health Organization. (2002). *National cancer control programmes : policies and managerial guidelines*, 2nd ed. World Health Organization. Available at: <https://iris.who.int/handle/10665/42494>

Worldometer, 2024. Nigeria Population. [online] Available at: <https://www.worldometers.info/world-population/nigeria-population/> Accessed 15th July 2024.

Zubairu, I.H. and Balogun, M.S., 2023. Population-based cancer registries in Nigeria and the National Cancer Control Programme. *ecancermedicalscience*, 17. Available at: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10550292/>